Express Riddler

22 April 2022

Riddle:

The Flash challenges Usain Bolt to a 100-meter race. Bolt runs at an average speed of 10 meters per second. To make it interesting, the Flash decides he will pick a random speed between 5 meters per second and 16 meters per second, with each speed in between being equally likely. (Note that fractional and decimal speeds are included here, rather than just whole numbers.)

On average, how often would you expect the Flash to win? What would be his average margin of victory?

Solution:

The Flash will win as long as his speed is greater than 10 m/s. As long as his speed is distributed evenly among real numbers, his speed will never equal exactly 10,and they will never tie. Then the probability P of his speed being greater than 10 is given by

$$P = \frac{\int_{10}^{16} dv}{\int_{5}^{16} dv}$$

which has the trivial solution of $^6/_{11} \approx 54.5\%$. The average margin of victory T is given by

$$T = \frac{\int_{10}^{16} \frac{100}{10} - \frac{100}{v}\, dv}{\int_{10}^{16} dv}$$

which has the solution $10 - {}^{50}/_{3} \ln({}^8/_5) \approx 2.17$. So the Flash will win (approximately) $\boxed{\textbf{54.5\%}}$ of the time, with an average margin of victory of (approximately) $\boxed{\textbf{2.17 s}}$.